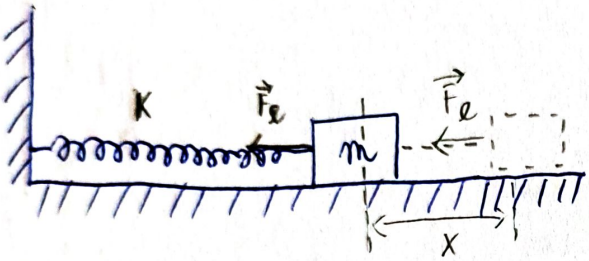
Fizică

Mișcare oscilatorie

1. Formalismul { newtonian / Lagrange / Hamilton
2. Soluția ec. dif. - oscilații libere (armonice liniare)
3. Oscilații amortizate - Mărimi caracteristice
4. Oscilații forțate - Fenomenul de rezonanță

Formalismul newtonian



P_2: $m\ddot{x} = -K_{el}x$

$\ddot{x} + \frac{K_{el}}{m}x = 0$

$\frac{K_{el}}{m} > 0 \Rightarrow \omega_0^2 \rightarrow$ pulsația mișcării

$\boxed{\ddot{x} + \omega_0^2 x = 0}$

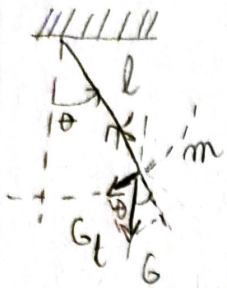
ec. dif. de ordin 2 liniară

Soluție

$x = e^{\lambda t}$

$\omega_0^2 = \frac{K_{el}x}{mx} = \frac{\text{forță de revenire}}{\text{masă} \cdot \text{deplasare}}$

(4)



$\approx$ opt. $\theta < 5^\circ$

$$\omega_0^2 = \frac{G_t}{m \cdot l(l\theta)} = \frac{mg\sin\theta}{ml\theta} \cong \frac{g}{l}$$

$$T = \frac{2\pi}{\omega_0}$$

oscilații izocrone!

Formalismul Lagrange

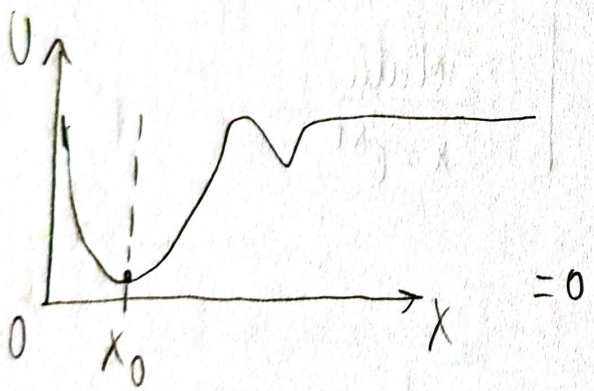
$$\mathcal{L} = \mathcal{L}(q_i(t), \dot{q}_i(t), t)$$

$$\downarrow$$

$$x(t)$$

$$\mathcal{L} = E_c - U = \frac{1}{2} m\dot{x}^2 - \frac{1}{2} k_{el} x^2$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right) = \frac{\partial \mathcal{L}}{\partial x} \qquad \frac{d}{dt}(m\dot{x}) = -k_{el} x \Rightarrow m\ddot{x} = -k_{el} x$$



$= 0$

$$\mathcal{U}(x) \cong U(x_0) + \frac{1}{1!}\left(\frac{dU}{dx}\right)_{x_0}(x - x_0) + \frac{1}{2!}\left(\frac{d^2U}{dx^2}\right)_{x_0}(x - x_0)^2 + \dots$$

$\mathcal{U}$t. $x_0 = 0$, $U(x_0) = 0 \Rightarrow$ $\mathcal{U}$t. $(x) \cong \frac{1}{2}\left(\frac{d^2U}{dx}\right)_0 x^2$

$$U = \frac{1}{2} k_{el} x^2 \Rightarrow |F| = -\frac{dU}{dx} = -k_{el} x$$

Formalismul Hamilton

$H = H(p, q) \rightarrow H = H(p, x)$

$H = E_c + U = \frac{p^2}{2m} + \frac{Kx^2}{2}$

$\frac{p^2}{2m} \leftarrow \frac{mv^2}{2} = \frac{m\dot{x}^2}{2}$

$$\begin{cases} \dot{x} = \frac{\partial H}{\partial p} \\ \dot{p} = \frac{\partial H}{\partial x} \end{cases} \longrightarrow \begin{cases} \dot{x} = \frac{p}{m} \\ \dot{p} = -Kx \end{cases} \longrightarrow \ddot{x} = \frac{\dot{p}}{m} = -\frac{Kx}{m}$$

$$\boxed{m\ddot{x} + Kx = 0}$$

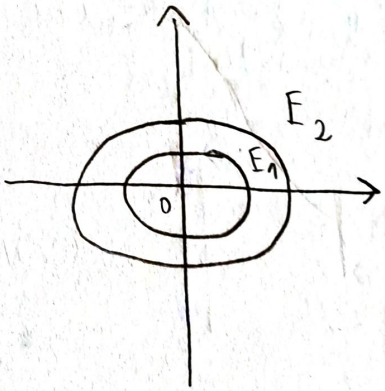
$$\frac{\dot{x}}{\dot{p}} = -\frac{1}{mKx} \Rightarrow \frac{\frac{\partial x}{\partial t}}{\frac{\partial p}{\partial t}} = -\frac{1}{x} \cdot \frac{1}{mK} \Rightarrow x\,dx = -\frac{1}{mK} p\,dp$$

$$\frac{x^2}{2} = -\frac{1}{mK} \cdot \frac{p^2}{2} + \text{const.}$$

$$\boxed{\frac{x^2}{2} + \frac{p^2}{2mK} = \text{const.}}$$

↓

elipsă



[redacted]

$\dot{x} = \lambda e^{\lambda t} \quad \ddot{x} = \lambda^2 e^{\lambda t}$

$(\lambda^2 + \omega_0^2) e^{\omega t} = 0$

$\boxed{\lambda^2 + \omega_0^2 = 0}$ ec. caracteristică

↓

$\boxed{\lambda_{1,2} = \pm i\omega_0}$

sol. exp.

$x(t) = A e^{i\omega_0 t} + B e^{i\omega_0 t}$

A și B se det. din condițiile inițiale: $\begin{cases} x(0) \\ \dot{x}(0) \end{cases}$

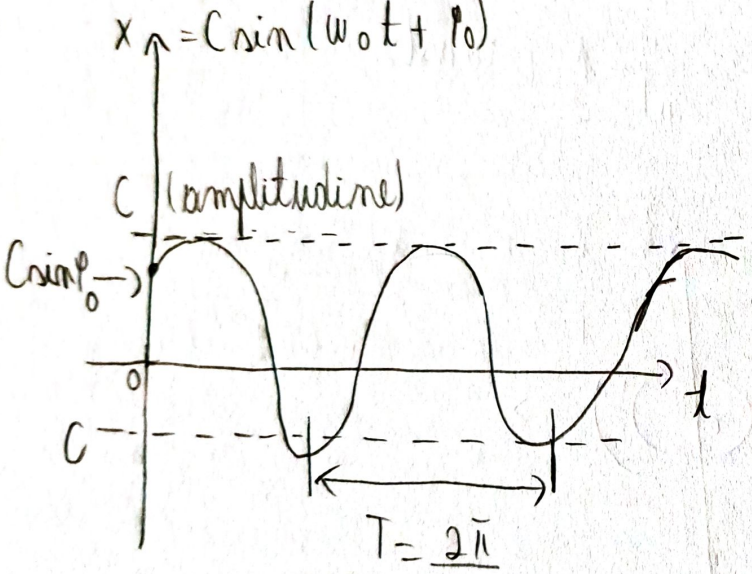
→ sol. trigonometrică

$x(t) = A_1 \sin(\omega_0 t) + B_1 \cos(\omega_0 t)$

→ sol. trigonometrică (2)

$x(t) = C \sin(\omega_0 t + \varphi_0)$ condiție inițială: φ_0

$x(t) = \tilde{C} e^{i\omega_0 t} = C(e^{i(\omega_0 t + \varphi_0)})$



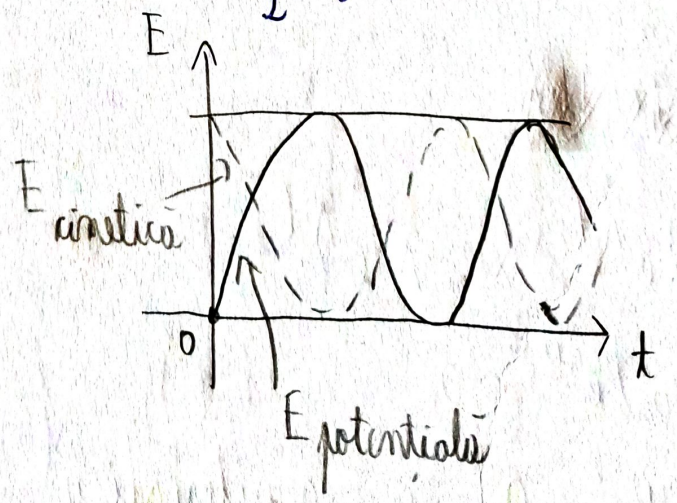
$\varphi(t) = \omega_0 t + \varphi_0$ FA2A

$E = E_c + U =$

$= \frac{m\dot{x}^2}{2} + \frac{K_{el} x^2}{2} =$

$= \frac{m}{2} C^2 \omega_0^2 [\underbrace{\cos^2(\omega_0 t + \varphi_0) + \sin^2(\omega_0 t + \varphi_0)}_{1}]$

$= \frac{K_{el} C^2}{2} = \text{const} = (E_c)_{max} = (U)_{max}$



$m\ddot{x} = -k_{el}x - \lambda\dot{x}$

λ const. F_f

$\ddot{x} + \frac{\lambda}{m}\dot{x} + \frac{k_{el}}{m}x = 0$

$\frac{\lambda}{m} \rightarrow 2\delta$ coef. de amortizare

$\frac{k_{el}}{m} \rightarrow \omega_0^2$ pulsație proprie

$$\ddot{x} + 2\delta\dot{x} + \omega_0^2 x = 0$$

sol. $x = e^{\lambda t}$

$(\lambda^2 + 2\delta\lambda + \omega_0^2)e^{\lambda t} = 0$

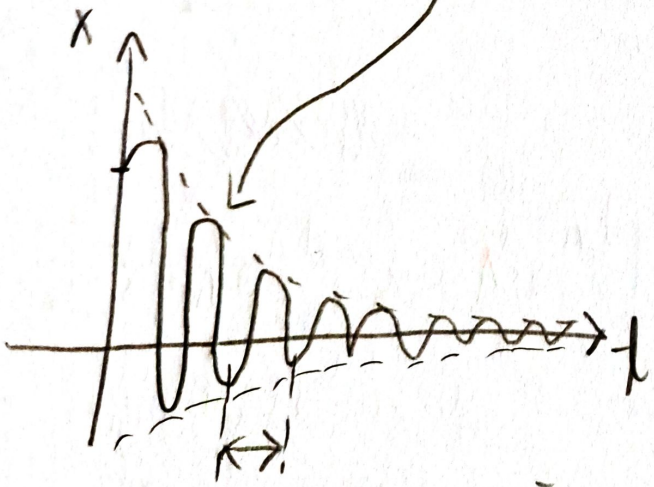
$\Rightarrow \lambda_{1,2} = -\delta \pm \sqrt{\delta^2 - \omega_0^2} \rightarrow \omega_1$

pt. $\delta^2 < \omega^2$ oscilații subamortizate

$\lambda_{1,2} = -\delta \pm i\sqrt{-\delta^2 + \omega_0^2}$ $\quad i^2 = j^2 = -1$

$\Rightarrow$ sol.: $x(t) = A e^{-\delta t} \cdot e^{i(\omega_0^2 - \delta^2)t} \rightarrow A e^{-\delta t} \cdot \sin(\omega_1 t + \varphi_1)$

$A e^{-\delta t} \rightarrow B = A e^{-\delta t}$ $\rightarrow$ scade exp. în timp



$$T_1 = \frac{2\pi}{\omega_1} = \frac{2\pi}{\sqrt{-\delta^2 + \omega_0^2}}$$

Mărimi caracteristice

a) timp de relaxare (τ_0)

$$\frac{Ae^{-\delta t}}{Ae^{-\delta(t+\tau_0)}} = e^{\delta\tau} = e^1 \Rightarrow \delta\tau = 1 \Rightarrow \boxed{\tau = \frac{1}{\delta}} \; !$$

b) decrement logaritmic (Δ)

$$\Delta = \ln \frac{Ae^{-\delta t}}{Ae^{-\delta(t+\tau)}} = \delta T_1$$

c) factor de calitate (Q)

$$Q = \pi N = \pi \frac{\tau}{T_1} = \frac{2\pi}{T_1} \cdot \frac{\tau}{2} = \frac{\omega_1}{2\delta} \; !$$

$N \rightarrow$ nr. de oscilații efectuat într-un timp de relaxare